

RESEARCH ARTICLE

2-Oxazoline-Based Polymer for Pharmaceutical Products Adsorption in Aqueous Media

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ABSTRACT

We present the synthesis and comprehensive evaluation of a novel 2-oxazoline-based polymer (polyPhOx) functionalized with phenyl groups, designed for the efficient adsorption of pharmaceutical contaminants from aqueous solutions. Employing living polymerization techniques (CROP) with two distinct initiators, we optimize reaction parameters including time, temperature, and initiator concentration. The rigorous characterization of the obtained polymer is depicted as well as the demonstration of its significant binding capacity for various pharmaceuticals, exploiting π - π and electrostatic interactions to achieve superior adsorption efficiency. Adsorption experiments are performed using high concentrations of model pharmaceuticals, including aspirin, ibuprofen, metoclopramide, pyramidone, oestradiol, and indomethacin, in hydroalcoholic solutions. Initial investigations confirm polyPhOx's exceptional efficacy, particularly with hydrophobic compounds such as oestradiol, achieving adsorption capacities of up to 56.67 mg g⁻¹. Further validation in real environmental samples highlights the polymer's practical applicability, showing substantial removal rates even under complex matrix conditions and bringing hope for effective water purification solutions. Remarkably, polyPhOx retains high adsorption performance after 10 cycles, underscoring its potential for sustainable and cost-effective water purification. This study not only introduces a promising material for environmental remediation but also demonstrates its robustness and reusability, paving the way for advanced wastewater treatment solutions targeting persistent pharmaceutical pollutants.

1 | Introduction

Nowadays the global consumption of organic compounds is undergoing a constant and rapid increase due to growing industrialization, new advanced agricultural practices, and technological expansions [1]. Moreover, the widespread consumption of pharmaceuticals has raised concomitant concerns about their presence in the environment, because a large proportion of these therapeutic compounds cannot be absorbed or metabolized by the human body. Thus, excreted via feces and urine [2], the major constituents of pharmaceutical waste,

including antibiotics, chemotherapeutics, hormones, analgesics, antipyretics, and antidepressants [3–5], enter the sewage system and eventually in water streams. It is already well established that some natural and synthetic therapeutics are currently under deep investigations for their ability to disrupt the normal functions of animals and humans [3]. For example, the incidence of endocrine-related diseases and adverse physiological effects in wildlife is increasing, and there are indications that changes in the reproductive health of humans, including declining male fertility, birth defects, and breast and testicular cancer, could be linked to the estrogen exposure [4]. Several recent studies

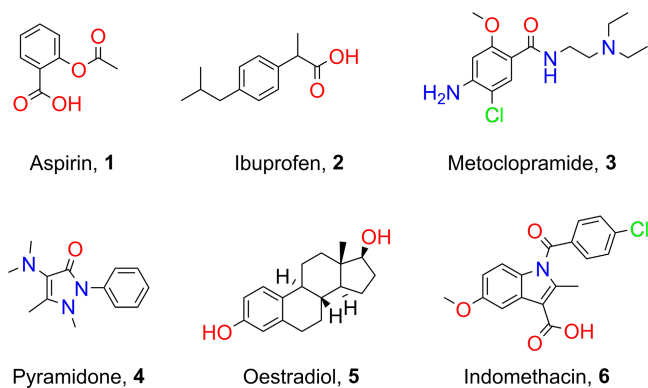


FIGURE 1 | Pharmaceutical compounds commonly detected in wastewater.

have drawn attention to the significant and alarming presence of certain common medicinal anti-inflammatory, analgesic, and steroid drugs (Figure 1), such as: aspirin (1), ibuprofen (2), metoclopramide (3), pyrimidone (4), oestradiol (5), and indomethacin (6) in the aquatic environment [4–6].

Such a dramatic scenario has motivated many governments and public health research institutions to focus their efforts on the monitoring of pharmaceutical residues and their effective removal from contaminated water source. The major removal mechanisms of these compounds in wastewater treatment plants (WWTP) using biological approaches are conventional activated sludge treatment, membrane Bio-Reactor, attached growth, constructed wetland, algae photobioreactor, and stabilization ponds [7]. Most common systems aim to eliminate soluble organic pollutants, suspended solids and flocculated matter resulting in high-quality effluent suitable for environmental discharge. Nevertheless, the current wastewater treatment systems are insufficiently effective or economical at removing persistent micropollutant due to their nature and lower quantity [8]. Additionally, these techniques are often impractical, as they incur high maintenance and operational costs, and often produce highly toxic secondary contaminants [8]. In this regard, our laboratory's previous endeavors provided colloiddally stable superparamagnetic nanoparticles (SPMNP) grafted with polyoxazoline-based polymers capable of efficiently interacting and selectively adsorbing heavy metal ions in aqueous solution [9]. Inspired by these findings, we envisage the possibility of removing organic pollutants, particularly pharmaceuticals, from wastewater using environmentally friendly poly(2-oxazoline)s (POxs) [10, 11]. Herein, we report the preparation of a new 2-oxazoline-based polymer with a novel side chain substituent bearing two aromatic rings, which introducing a new functionality through its coplanar structure may allow π - π interactions with aromatic compounds. The adsorption capacity of the resulting POx is validated upon several experiments employing the titled pharmaceuticals dissolved in different aqueous media.

2 | Results and Discussion

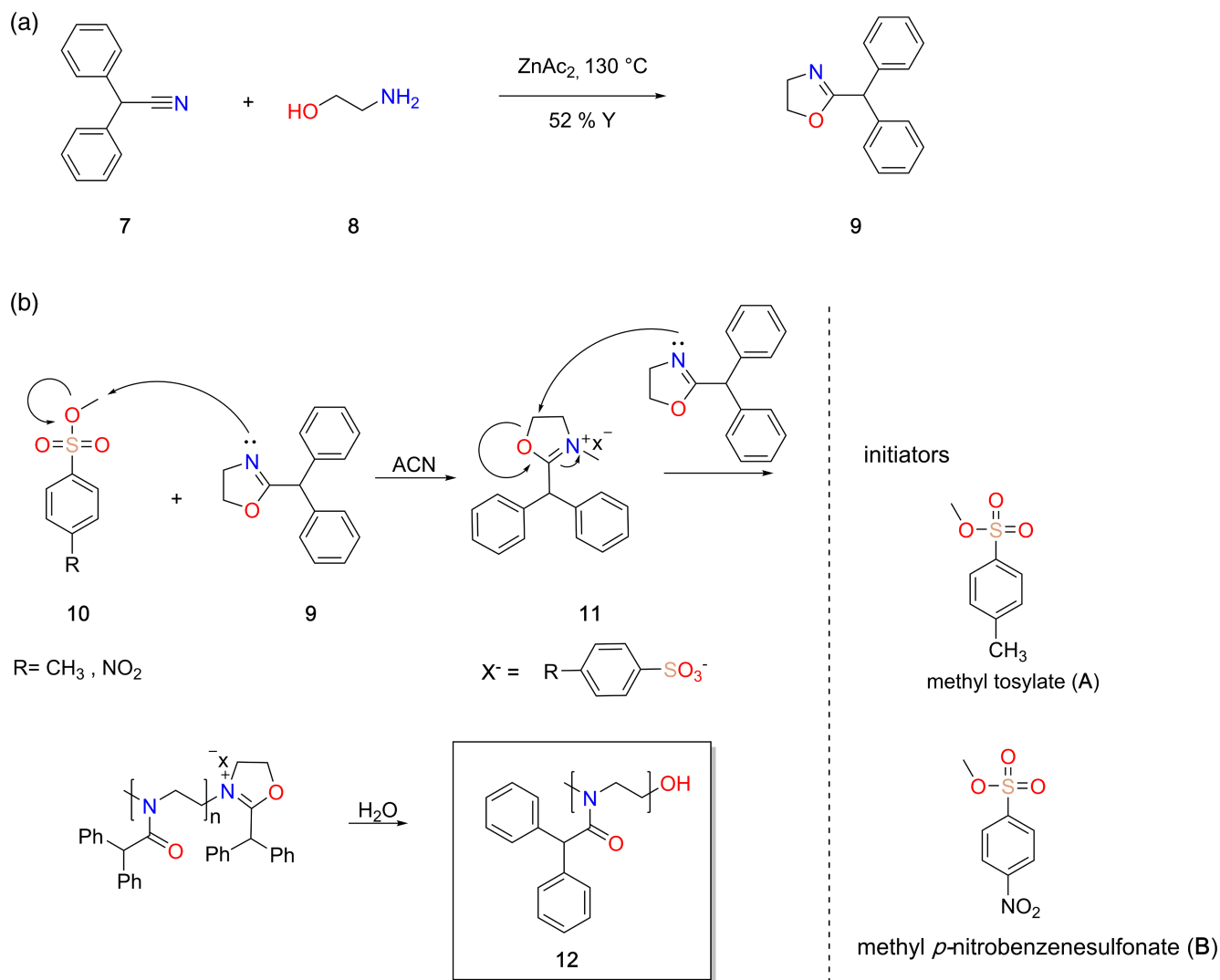
2.1 | Preparation of the Adsorbent

In the last 50 years a wide range of POxs have been straightforwardly prepared through Cationic Ring-Opening Polymerization (CROP) [10] employing variously decorated 2-oxazolines as suitable cyclic

imino ether monomers [12, 13]. By circumventing the formation of termination products and inhibiting the chain transfer, the selected living polymerization enables a precise regulation of the polymer chain length by merely adjusting the reaction conditions [10, 14–16]. According to this approach, we initially synthesized the 2-benzhydryl-4,5-dihydro-1,3-oxazole **9** following a modified Witte-Seeliger cyclocondensation promoted by a metal catalysis (Scheme 1). Specifically, the desired monomer **9** was isolated in fairly good yields treating the 2,2-diphenylacetone nitrile (**7**, 1 equiv.) with 2-aminoethanol (**8**, 1.3 equiv.), in presence of zinc acetate hydrate (0.02 equiv.) as catalyst, and purifying the resulting crude reaction mixture with column chromatography (Scheme 1a) [17, 18]. After the monomer characterization through NMR, FT-IR, GC-MS, and HRMS, the polymerization was hence performed by adding the initiator to a solution of 2-benzhydryl-4,5-dihydro-1,3-oxazole (**9**, 10 mmol) in anhydrous acetonitrile (ACN) (5 mL). Mechanistically, CROP is initiated by generating an oxazolinium cation **11** that immediately reacts with the nucleophilic nitrogen atom of another monomer **9**. The living chain-ends are terminated intentionally by the addition of a nucleophile stronger than the monomer (Scheme 1b).

Taking into account the just depicted mechanism, different reaction conditions (i.e., type and concentration of the initiator, and temperature) were tested in order to select the most appropriate in terms of both yield (*Y*) and polydispersity index (*D*). The most significant tests conducted together with the corresponding results, are shown in Table 1.

Although the dimethylacetamide (DMA) and chlorobenzene, usually involved in CROP, produced comparable results, acetonitrile (ACN) was used as solvent of choice due to its less toxicity, high availability, and ease of use. Since the methyl tosylate (**A**) normally furnished well defined polymers with narrow molar mass distribution [19, 20], we initially tested the polymerization reaction using the freshly prepared 2-oxazoline **9** and different amounts of **A**. While the lowest temperature did not allow the formation of polyPhOx **12** except in traces and after an excessively long reaction time (entry 2), performing the polymerization at 90°C with 0.01 equiv. of **A** for only 24 h provided the desired product **12** in very good yield (72%, entry 4), above all when 0.02 equiv. of **A** was added (95%, entry 3). A reduction in the reaction time from 24 to 8 h (entry 5) resulted in a slight decrease in yield from 95% to 85%, as well as in a reduction of the molecular weight (from 24.6 to 19.7 kDa); while this approach led to an enhancement in monodispersity (*D* = 1.5). Moreover, in contrast with the results reported for other alkyl 2-oxazolines [19, 21, 22], the GPC analysis revealed globally a too high heterogeneity of the samples (polydispersity index, *D* = 2.8) even when we reduced both the initiator amount (*D* = 1.9, entry 4) and the reaction time at 8 h (*D* = 1.5, entry 5) employing methyl tosylate (**A**) as initiator. Thus, we decided to screen other sulfonates as initiator and, among others, the best results were observed with *p*-nitrobenzenesulfonate (**B**). Indeed, the desired polymer was isolated in good-to-high yield (75%–88%, entries 6 and 7), reasonable reaction time (8 h) and, mostly important, with a narrow polydispersity index (*D* = 1.1). Such successful outcome is probably due to the higher leaving group ability of the *p*-nitrobenzenesulfonate (**B**) as initiator which leads to a quantitative initiation step faster than the propagation one. This results in a uniform growth of the polymer chains, leading a



SCHEME 1 | (a) 2-Benzhydryl-4,5-dihydro-1,3-oxazole **9** synthesis; (b) CROP mechanism for poly(2-benzhydryl-4,5-dihydro-1,3-oxazole) (polyPhOx **12**).

TABLE 1 | Polymerization reactions with methyl tosylate (**A**) and methyl *p*-nitrobenzenesulfonate (**B**) as initiators.

Entry	Initiator	M_R^a (eq.)	R_t^b	Temp (°C)	$M_w^{c,f}$	$D^{d,f}$	Y^e
1	A	0.02	5 d	50	N.D.	—	—
2	A	0.01	11 d	25	1.4	—	4%
3	A	0.02	24 h	90	24.6	2.8	95%
4	A	0.01	24 h	90	34.7	1.9	72%
5	A	0.02	8 h	90	19.7	1.5	85%
6	B	0.02	8 h	90	6.5	1.1	88%
7	B	0.01	8 h	90	6.8	1.1	75%

^a M_R indicates the molar ratio.

^b R_t indicates retention time.

^c M_w indicates molecular weight (kDa).

^d D indicates polydispersity index.

^e Y indicates the observed yields calculated.

^f M_w and D were obtained with GPC analyses using polystyrene reference materials and DMF 50 mM LiBr as eluent.

lower D value with **B** as initiator with respect to **A**. In fact, the propagation rate of this reaction is supposed to be very high due to the presence of an aromatic substituent that could increase

the electronic donation effect, stabilizing the intermediate cation **11** (Scheme 1a). The polymer obtained by employing 0.02 equiv. of **B** and carrying out the reaction at 90°C for 8 h (entry

6) represents, at this stage of our investigation, the optimal compromise among length, number of monomers, and polydispersity. This should enable us to gain a better and deeper understanding of the intricate adsorption process performed by the polymer itself in aqueous media. Thus, once the desired polymer was fully characterized with FT-IR and thermogravimetric analyses (TGA) to evaluate the thermal stability of the polymer, and it was produced on a gram scale, polyPhOx **12** was subjected to a series of tests to validate the adsorption efficiency of pharmaceuticals in water.

2.2 | Adsorption Experiments

Aspirin (**1**), ibuprofen (**2**), metoclopramide (**3**), pyramidone (**4**), oestradiol (**5**), and indomethacin (**6**) were selected as model organic pollutants (Figure 1) due to their well-documented chemical and physical properties and their widespread presence in municipalized wastewater [23–25], above all in livestock farms, slaughterhouses, and large urban agglomerations [26].

2.2.1 | Evaluation of Adsorption Under Batch Conditions

Initially, adsorption experiments were performed under batch operation conditions in water:methanol (70:30) to ensure

homogeneous solutions and to achieve reliable and reproducible results even at the highest concentration of the most non-polar compound. Specifically, 2 mL of pharmaceuticals in hydroalcoholic solution at different concentrations were added to 5 mg of polymer keeping the resulting system under stirring for 24 h at room temperature. For each pharmaceutical, the adsorption capacity (q_e [mg g⁻¹]) and the percentage removal efficiency ($R\%$) were evaluated comparing the amount of adsorbed pollutant to either the amount of the employed adsorbent (Equation 1) or to the initial concentration of the drug (Equation 2):

$$q_e = \frac{(C_0 - C_e)V}{W} \quad (1)$$

$$R\% = \frac{(C_0 - C_e)}{C_0} \times 100 \quad (2)$$

where, C_e (mg L⁻¹) is the experimental liquid-phase concentration of pharmaceutical at equilibrium, C_0 (mg L⁻¹) is the average concentration of pharmaceutical in the corresponding control, V is the volume of the solution (L), and W is the amount (g) of the adsorbent used. Figure 2a,b provided clear evidence of a quite linear trend in the adsorption coefficient (q_e) as a function of the initial concentration of the tested pharmaceuticals up to 200 mg L⁻¹, a concentration that largely exceeds the legal limits. It is noteworthy that polyPhOx (**12**) demonstrated remarkable

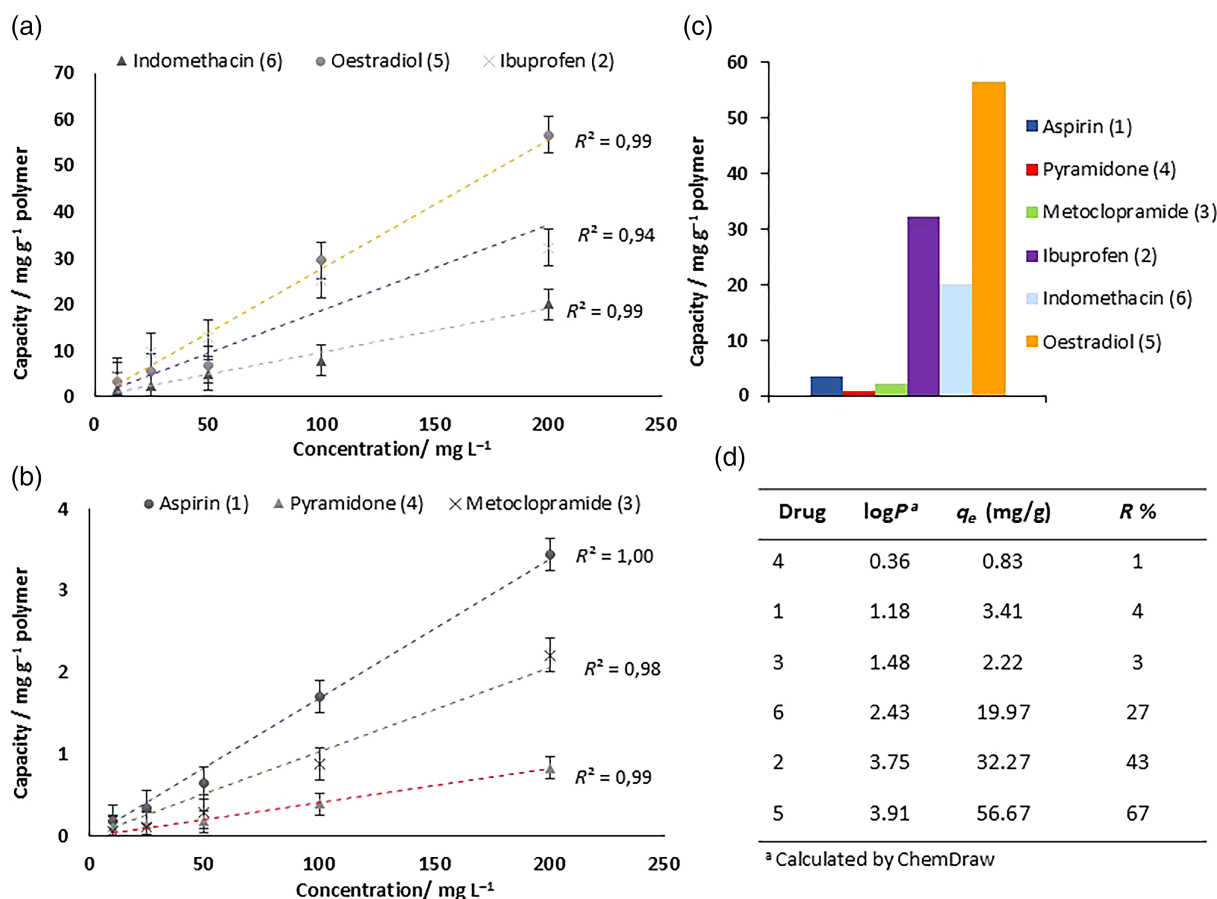


FIGURE 2 | (a) Adsorption capacity (q_e) versus initial concentration for **6**, **5**, and **2**. (b) q_e versus initial concentration for **1**, **4**, and **3**. (c) q_e for pharmaceuticals at 200 mg L⁻¹ as initial concentration. (d) Table of log P values and removal % ($R\%$).

efficacy in the removal of ibuprofen (**2**) and indomethacin (**6**) for which we found adsorption capacities of 32.27 and 19.97 mg g⁻¹, respectively. Furthermore, it exhibited excellent performance in the removal of oestradiol (**5**) with a q_e of 56.67 mg g⁻¹ (Figure 2c). In fact, oestradiol (**5**) has the lowest water solubility being the most hydrophobic compound, and this allows it to be the most suitable to interact with the polymer hydrophobic domains. The results presented above indicate that the polyPhOx **12** treatment was markedly more effective in removing those ligands with lower water solubility offering a hopeful prospect for future water treatment methods. A comparison of the removal percentage ($R\%$) with the solubility in water ($\log P$), reveals that the highest value of $R\%$ was observed for those ligands with higher $\log P$ (Figure 2d). This suggests that non-polar species have a low affinity for water and aqueous environments, thus favoring absorption within the hydrophobic domains of the polymer structure. Such a behavior is probably due to the formation of several π - π and electrostatic interactions between the hydrophobic part of the pollutants and the aromatic substituents on the polymer chains. In line with such considerations, for aspirin (**1**), metoclopramide (**3**), and pyrimidone (**4**) we found lower values of adsorption capacity, that is 3.41, 2.22, and 0.83 mg g⁻¹, respectively. Indeed, compounds **1**, **3**, and **4** appear to have a low ratio between hydrophobic and hydrophilic moieties with consequently significantly lower $\log P$ values (i.e., 1.18, 1.48, and 0.36, respectively). In this regard, the stabilization of the aqueous media has the effect of lowering the affinity for the hydrophobic phase, despite of the presence of the aromatic rings and the possible favorable π - π stacking. Also, the steric hindrance may contribute to prevent the formation of stable interactions between the aromatic groups of polyPhOx **12** and those of the tested pharmaceuticals. In order to study the distribution of the adsorbed drugs between the liquid and the adsorbent phase, the equilibrium adsorption capacity data were analyzed according to the Langmuir and Freundlich models of adsorption isotherms. The thermodynamic parameters calculated using the latter model were in good agreement with the experimental results, confirming the hypothesis of multilayer adsorption on heterogeneous sites.

2.2.2 | Evaluation of Adsorption in Real Matrixes

The adsorption performance of the prepared polymer was subsequently evaluated using a water sample from Albano Lake, which was selected as the model for a more complex environmental matrix. Pharmaceuticals were spiked into the lake water (pH 8.20), which serves as free control sample at a concentration of 10 mg L⁻¹. This concentration represents the best compromise in terms of solubility for the most hydrophobic compound (oestradiol (**5**)) and the experimental reproducibility. Removal tests on these water samples were carried out in triplicate using the same protocol previously described. As depicted in Figure 3a, it can be observed that pyrimidone (**4**) and metoclopramide (**3**) (i.e., the most hydrophilic compounds with $\log P = 0.36$ and 0.80), were hardly adsorbed (1%). The removal percentages of aspirin (**1**), indomethacin (**6**), and ibuprofen (**2**) significantly decreased with respect of the hydroalcoholic solution, moving to 1% from 4%, 27%, and 43%, respectively. This phenomenon is likely attributable to the acid-base reactions that occur at a pH of 8.20. Indeed, since these compounds have pKa values of 3.26, 2.99, and 4.37 respectively, they are more stable in the deprotonated form within this matrix, thereby preventing them from being effectively adsorbed by the polymer. Differently, oestradiol (**5**), the most hydrophobic compound ($\log P = 3.91$) was removed with the greatest efficiency (60%) by the tested polymer. This suggests that our system could be very attractive for the selective removal of non-polar pollutants, particularly this specific pollutant, in real matrixes. For aspirin (**1**) which has a good compromise in terms of solubility in water and removal percentage in the hydroalcoholic experiments ($\log P = 1.18$ and $R\% = 4$), we also performed experiments in spiked Albano Lake water at various concentrations to observe any growth of the q_e , despite the removal percentage in this matrix is very low (1%). Figure 3b illustrates a linear trend between the adsorption coefficient and the initial concentration of the drug ($R^2 = 0.98$), although this value is lower than that obtained from the experiments conducted in the hydroalcoholic solutions ($q_e = 2.25$ instead of 3.41). It is postulated that this discrepancy may be attributed to the deprotonation effect previously discussed.

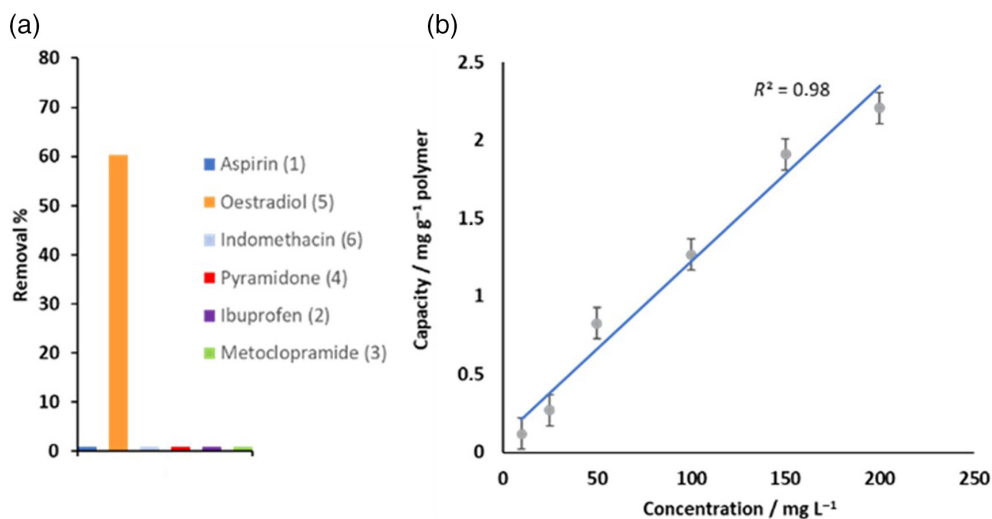


FIGURE 3 | (a) Removal percentage of selected drugs in Albano Lake water for pharmaceuticals at a concentration of 10 mg L⁻¹; (b) adsorption capacity (q_e) versus initial concentration for ASA (**1**) in spiked Albano Lake at various concentrations.

2.2.3 | Reusability

The reusability of adsorbents is a crucial characteristic that should be taken into account when assessing the potential applicability of new materials. However, this aspect is frequently overlooked during the development process. To be reused, the adsorbent should successfully release the contaminant by a desorption or regeneration step, without compromising the chemical and physical stability of the matrix [27]. After each adsorption experiment at the maximum initial concentration of the drug (200 mg L^{-1}), the polymer (5 mg) was subjected to a series of washes with tap water (15 mL, three times), with water (15 mL, three times) and finally with methanol (15 mL). The adsorption experiment was then repeated with the same material and the q_e was calculated for each drug. Only after the completion of 10 cycles, a slight decrease in adsorption capacity (q_e) was noted, amounting to a 1% reduction (Figure 4).

3 | Conclusions

A novel 2-oxazoline-based monomer was synthesized in high yields, fully characterized, and subsequently subjected to the cationic ring-opening polymerization. The newly obtained polymer (polyPhOx, **12**) was straightforwardly employed for the removal of pharmaceutical compounds from aqueous solutions. The observed positive outcomes highlighted the remarkable efficiency of the evaluated systems, especially in the adsorption of the most apolar organic compounds (i.e., oestradiol (**5**), ibuprofen (**2**), and indomethacin (**6**)), which demonstrate higher affinity for the fixed polymeric phase. Subsequently, several tests were also carried out using real water samples, taken from Albano Lake. It is noteworthy that the performed analyses demonstrate the good effectiveness and high selectivity in oestradiol (**5**) removal from both hydroalcoholic solutions and complex aqueous media (water from Albano Lake), even when the majority of tested compounds

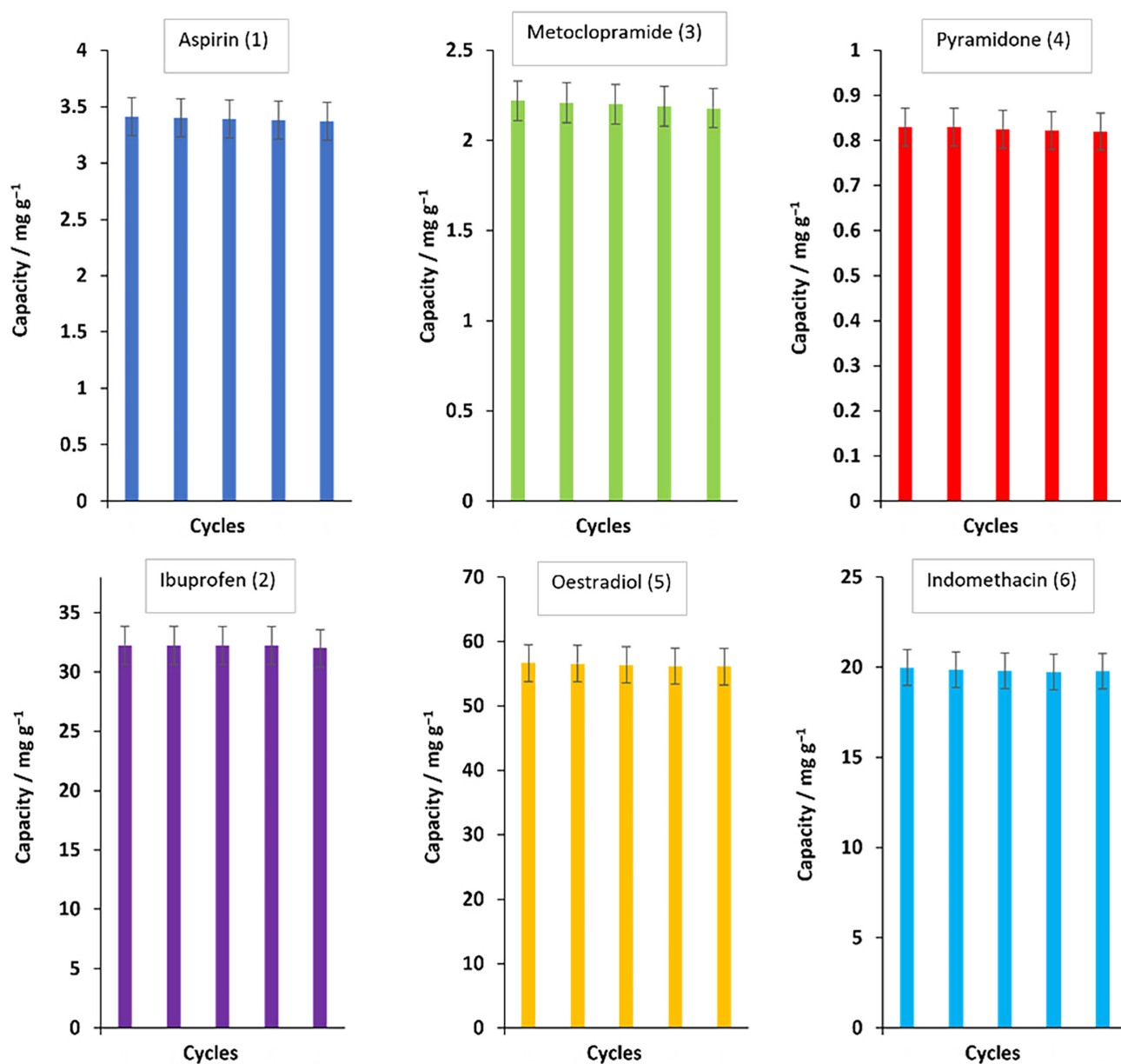


FIGURE 4 | q_e Calculated after every two loading cycles for the target pharmaceutical compounds.

are minimally adsorbed, because of the deprotonation effect. Such results, that is, experiments conducted in a more realistic pollution scenario, provide a promising avenue for further exploitation of the adsorptive properties of our polymers, offering a valuable alternative to existing systems. Moreover, although polyPhOx **12** belongs to a class of environmentally friendly polymers, several experiments are ongoing to evaluate the metabolic pathways and cytotoxicity of both the polymer and its degradation products. Likewise, research is already underway to investigate the performance of the developed system under various operating conditions including flow, temperature variation, atmospheric condition, and pH variation. It should be noted that this study is not intended to be conclusive, as there are still open issues related to the synthesis of 2-oxazolines featured with additional structural and electronic modifications on the aromatic moieties, and to the improvement of the final M_w without compromising the observed monodispersity. Furthermore, in addition to the ongoing computational studies on PiPOx, we intend to supplement the present study on the newly synthesized polyPhOx by employing computational modeling [28] to gain deeper insights into the underlying mechanism governing its adsorption selectivity at the molecular level. Likewise, in a more applicative perspective, future investigations will address in-depth considerations in terms of target organic pollutants, assessment on urban and industrial wastewater samples.

4 | Experimental Section

All reactions were monitored by thin layer chromatography (TLC) carried out on Merck silica gel 60 F254 (0.04–0.063 mm) plates and visualized with UV light (254 nm). Flash chromatography was performed on SigmaAldrich silica gel (particle size: 0.040–0.063 mm). ^1H NMR and ^{13}C NMR were recorded in CDCl_3 (99.8% in deuterium) using a Bruker DRX-600 spectrometer (600 MHz, Billerica, MA, USA). All chemical shifts are expressed in parts per million (δ scale) and are referenced to the residual protons of the NMR solvent (CDCl_3 , δ 7.26 ppm). The following abbreviation was used to explain the multiplicities: (s) = singlet, (d) = doublet, (t) = triplet, (m) = multiplet; coupling constants (J) were expressed in Hz. Infrared Spectra (FT-IR) were obtained using a Nicolet 6700 FT-IR Spectrometer from Thermo Fisher in KBr pellets, 64 scan, resolution 4 cm^{-1} data are presented as the frequency of absorption (cm^{-1}). Melting points were determined with a Mel-Temp. HRMS Spectra were recorded with micromass Q-TOF micromass spectrometer (Waters). Micromass LCT (ESI) with Lock-Spray-Injector (Injection Loop-Modus in a HPLC system, Waters, Alliance 2695). Reversed phase High Performance Liquid Chromatography (HPLC) were performed with an Agilent infinity 1260 II instrument equipped with a Luna 5u C18(2) 100 A Column ($5\text{ }\mu\text{m}$, $100\text{ }\text{\AA}$, $150\times 4.6\text{ mm}$, Phenomenex) with DAD as detection, and with water 1 mM HClO_4 (A) and acetonitrile (B) as mobile phases. Gradient conditions (with 1.00 mL min^{-1} flow) were: initial composition 20% B to 100% B over 20 min; then held for 10 min at 100% B and finally returned to 20% B over 2 min. Thermogravimetric analyses (TGA/DTG) were performed using a SDT Q600 thermoanalyzer (TA Instruments). Aliquots of 20.0 mg of each sample at dry state, were inserted into a platinum crucible and heated from room temperature ($25\pm 1^\circ\text{C}$) to

800°C with a heating ramp of 10 K min^{-1} under nitrogen flow (100 mL min^{-1}). Gel Permeation Chromatography (GPC) setup consists of three columns in series (a precolumn followed by two columns [$300\times 4.6\text{ mm}$] $5\text{ }\mu\text{m}$ 10 E5 $\text{\AA}$ [$10\text{--}1.000\text{ K}$] and $5\text{ }\mu\text{m}$ 10 E3 $\text{\AA}$ [$1\text{--}75\text{ K}$] respectively). Samples were injected with a concentration of 1.0 mg mL^{-1} and measured at 25°C using DMF 50 mM LiBr as eluent, 0.27 mL min^{-1} flow and DAD as detector. The calibration was conducted with standard polystyrene reference materials (M_w ranged from 1.3 to 900 K). GC-MS analyses were performed with a SHIMADZU GC-2010 with a capillary column of SLBTM-5ms ($30\text{ m}\times 0.25\text{ mm}\times 0.25\text{ }\mu\text{m}$) coupled with a Mass Spectrometer SHIMADZU QP 2010S; the mass values are reported as mass/charge ratios (m/z). All analyses were performed with a column temperature program from 50°C to 250°C ($10^\circ\text{C min}^{-1}$) and the injection temperature was set to 250°C .

4.1 | Starting Materials

Solvents and common reagents were purchased from commercial sources without any further purification (Sigma-Aldrich). Methyl tosylate (98%) was purified by distillation at reduced pressure and stored under Argon atmosphere. Acetonitrile (ACN) (HPLC gradient grade) was distilled over calcium hydride (CaH_2) and stored under Argon atmosphere.

4.2 | Synthesis of 2-Benzhydryl-4,5-Dihydro-1,3-Oxazole (9)

2-Benzhydryl-4,5-dihydro-1,3-oxazole (**9**) was synthesized by a modified Witte-Seeliger cyclocondensation of 2,2-diphenylacetone nitrile (**7**) (1 equiv.) with 2-aminoethanol (**8**) (1.3 equiv.), and zinc acetate hydrate (0.02 equiv.) as catalyst. The reaction mixture was kept under stirring at 130°C for 48 h with an Argon flow passing through the reaction vessel. The reaction was followed with TLC using petroleum ether:ethyl acetate (8:2) as eluent. Substrate conversion was around 55% after 48 h determined by GC-MS. The crude product was dissolved in dichloromethane (DCM) and extracted several times with water. The combined organic layers were washed with brine, dried with anhydrous magnesium sulfate (MgSO_4), and concentrated under vacuum. The residue was purified by flash chromatography on silica gel using petroleum ether:ethyl acetate (8:2) as eluent. The desired monomer was obtained as a white powder in 52% yield. m.p. $103^\circ\text{C}\text{--}105^\circ\text{C}$. GC-MS Retention time: 23.5 min, $m/z = 236$ base peak, $m/z = 237$ [M^+], FT-IR spectrum (KBr pellet): $\tilde{\nu} = 3059\text{--}2873$, 1658, 1492, 1150 cm^{-1} . ^1H NMR (600 MHz, CDCl_3) $\delta = 3.92$ (t, $J = 9.5\text{ Hz}$, 2H, $\text{OCH}_2\text{CH}_2\text{N}$), 4.30 (t, $J = 9.5\text{ Hz}$, 2H, $\text{OCH}_2\text{CH}_2\text{N}$), 5.11 (s, 1H, CCHPh_2), 7.33 (m, 10H, CH_{arom}) ppm. ^{13}C NMR (150 MHz, CDCl_3) $\delta = 168.4$, 139.6 (2C), 129.1 (4C), 127.3 (6C), 67.9, 54.7, 51.3 ppm. HRMS: exact mass calculated for ($\text{C}_{16}\text{H}_{15}\text{NNaO}$) requires m/z 260.1051, found m/z 260.1053.

4.3 | Synthesis of Poly(2-Benzhydryl-4,5-Dihydro-1,3-Oxazole) (12)

A two-neck round bottom flask equipped with an efficient reflux condenser and a magnetic stirring bar was charged with

2-benzhydryl-4,5-dihydro-1,3-oxazole **9** (10 mmol) and anhydrous ACN (5 mL). The system was kept under Argon atmosphere and a proper amount of initiator was added. The reaction was quenched by adding a solution of NaOH 1 M (excess to the initiator) and the resulting mixture was stirred for 2 h. The solvent was evaporated under vacuum and the residue was added to a solution of petroleum ether:diethyl ether (1:1). The polymer was recovered as a white precipitate, the supernatant was removed, and the product was washed twice with the petroleum ether:diethyl ether solution and dried in vacuo. m.p. 263°C–265°C. FT-IR analysis (KBr pellet): $\tilde{\nu}$ = 3447, 3066–2850, 1646, 1493–1417, 1164 cm⁻¹.

4.4 | Adsorption Experiments

Adsorption experiments were done in triplicate under batch operation conditions targeting pharmaceutical compounds. First, an amount of 5 mg of the polymer was added into 2 mL reaction tubes. Then, 2 mL of standard solutions with concentrations ranging from 10 to 200 mg L⁻¹ were loaded into the cartridge. To allow homogenous adsorption on the polymer surface, the tubes were placed in a thermostatic shaker and incubated at room temperature at 50 rpm for 24 h. Triplicate controls, which consisted of solutions of the corresponding pharmaceutical in the absence of adsorbent and in the same conditions, were run in parallel. Finally, the supernatants were filtered using PTFE filters, transferred into glass vials, and analyzed by HPLC-DAD. Adsorption experiments in Albano Lake water were conducted in triplicate as previously described: an amount of 5 mg of the polymer was added into 2 mL reaction tubes. Then, 2 mL of standard solutions with concentrations of 10 mg L⁻¹ (for aspirin (**1**)) concentration ranged from 10 to 200 mg L⁻¹ were loaded into the cartridge. To allow homogenous adsorption on the polymer surface, the tubes were placed in a thermostatic shaker and incubated at room temperature at 50 rpm for 24 h. Triplicate controls, which consisted of solutions of the corresponding pharmaceutical in the absence of adsorbent and in the same conditions, were run in parallel.

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Supporting Information

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